

A Resolution Framework for Finite-Sample Model Selection

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Abstract

The ML community selects models using finite validation sets, often at budgets where classical statistics predicts that rankings cannot be resolved. This paper proposes a pre-experimental diagnostic protocol to determine whether the evaluation budget can support statistically distinguishable comparisons. The protocol centers on $\Delta_{\min}$, a resolution threshold derived from the two-proportion z -test. It embeds a formal decision rule (Decision Rule 1): when the observed performance range falls below $\Delta_{\min}$, uniform model averaging (soup) minimizes expected regret; when it exceeds $\Delta_{\min}$, argmax is preferable. We validate the protocol on Qwen2.5-7B and Mistral-7B across multiple seeds, scales, and training paradigms. At full GSM8K ($N = 1319$), $\Delta_{\min} = 3.8$ pp exceeds the observed checkpoint range of 3.0 pp. Crucially, the protocol does not uniformly predict failure: for Mistral-7B at $N = 200$, $\Delta_{\min} = 8.11$ pp is exceeded by the 10.5 pp observed range. The decision rule selects soup across all three Qwen seeds (consistent) and argmax for Mistral (consistent). Bootstrap, Monte Carlo calibration, and continuous-metric extension (cross-entropy loss variance 0.18, $\Delta_{\min} \approx 0.06$) further support the framework. A quick-reference table and a catalog of nine pitfalls accompany the protocol; an exploratory survey of 36 papers (78 claims) finds that 71% of claimed improvements fall below $\Delta_{\min}$.

Keywords: Model selection, evaluation budget, statistical resolution, $\Delta_{\min}$, fine-tuning, reproducibility

1 Introduction

Model selection in machine learning follows a default rule: evaluate each candidate on a held-out validation set and pick the one with the best metric. This rule assumes that a model scoring 55% is genuinely better than one scoring 52% on the same test set. The assumption is known to be fragile (Kaur et al. 2025), but the source of the fragility is debated. Is ranking failure a rare event triggered by distribution shift or overfitting? Or is it a systematic limitation of finite-budget evaluation?

We argue that it is the latter. Model ranking is constrained by the evaluation budget, much as any statistical estimator is constrained by sample size. Existing comparison tools—paired t -tests (Dietterich 1998), significance guidelines (Dror et al. 2018), post-selection inference (Zrnic and Fithian 2024; Zhang et al. 2024)—are

post-hoc: they assess results after evaluation is complete. What is missing is a *pre-experimental diagnostic* that determines, before resources are invested, whether the evaluation budget can support statistically distinguishable comparisons.

This paper proposes such a diagnostic. It centers on $\Delta_{\min}$, a resolution threshold derived from the two-proportion z -test (Cohen 1988). The threshold itself is a textbook formula. The contribution is the framework built around it: a decision rule (Decision Rule 1) that formalizes when to use argmax versus model averaging, a calibration procedure via bootstrap and power-curve simulation, empirical validation across two model families and multiple training paradigms, a catalog of diagnostic pitfalls encountered during the research process, and a literature survey documenting the gap between what budgets provide and what practitioners assume. The paper identifies and quantifies a systematic mismatch: the ML community routinely operates at evaluation budgets where the resolution is insufficient to support the comparisons being drawn.

Contributions

1. **A resolution framework for model selection.** We introduce $\Delta_{\min}$, derived from classical power analysis, which captures the minimum difference between two models that reaches nominal significance at a given evaluation budget N . When the observed range falls below $\Delta_{\min}$, ranking is not statistically resolvable from this budget alone.
2. **Empirical validation across budgets and domains.** We validate the framework through controlled LoRA fine-tuning experiments (two model families: Qwen2.5-7B and Mistral-7B, 1.5B scale, full fine-tuning, multiple random seeds), starting with the full GSM8K test set ($N = 1319$) and tracing how resolution degrades under subsampling. The Mistral-7B experiment provides a discriminative check: $\Delta_{\min}$ correctly identifies one configuration as resolvable and another as not. The framework is not uniformly biased. The pattern is further replicated on classical ML model selection (random forests on MNIST) and synthetic simulations.
3. **A diagnostic failure mode catalog.** We document nine diagnostic pitfalls in empirical model evaluation, grouped into three categories, each with concrete detection and prevention strategies.
4. **A validated decision rule.** We propose a soup-vs-argmax heuristic (Decision Rule 1) and show empirically that its output is stable across three training seeds and two model architectures (Qwen2.5-7B and Mistral-7B).

2 Background and Related Work

2.1 The Model Selection Problem

Model training produces a sequence of candidates $c_1, \dots, c_T$ from which one must be chosen. The standard selection rule is to evaluate each on a held-out validation set and pick the one with the best metric. The implicit assumption is that the validation metric reliably indicates deployment quality.

2.2 Statistical Comparison of Models

Comparing models on finite evaluation samples is a known problem. Dietterich (1998) introduced the $5 \times 2cv$ paired t -test to address the high variance of single-sample comparisons. Demšar (2006) extended this to multiple classifier comparison. In NLP, Card et al. (2020) showed that underpowered test sets are common, and Dror et al. (2018) provided significance testing guidelines.

Our work differs in two respects. First, we focus on comparisons within a single training trajectory where the same test set is reused, creating a paired structure

that standard cross-model comparisons do not address. Second, we treat evaluation resolution as a pre-experimental diagnostic rather than a post-hoc test.

Nguyen et al. (2025) proposes uncertainty-guided checkpoint selection for reinforcement fine-tuning, using prediction uncertainty to identify the most promising candidate. Their approach addresses the question of *which* checkpoint to select given uncertainty estimates. $\Delta_{\min}$ addresses a prior question: *whether* the evaluation budget supports any reliable selection at all. We applied the principle of uncertainty-guided selection to our Qwen2.5-7B bootstrap distribution at $N = 200$. The selection entropy is 0.717 and the best model is chosen only 44% of the time. Even with an optimal uncertainty-aware rule, the fundamental ambiguity at this budget cannot be resolved. The two approaches are complementary. $\Delta_{\min}$ determines whether selection is feasible. Uncertainty-guided methods choose among candidates once feasibility is confirmed.

Shihab et al. (2025) detects proxy gaming. Ramé et al. (2024) averages reward model weights. Xu et al. (2026a) treats checkpoint selection as a robust decision problem under evaluation uncertainty. These are complementary to our diagnostic framing.

Several recent papers address the statistical challenge of ranking under evaluation noise. Zhang et al. (2024) develop confidence sets for the argmin (best model) from noisy observations, using cross-validation with an exponential weighting mechanism borrowed from differential privacy to handle the double-dipping bias in post-selection inference. Adrian et al. (2024) stabilize black-box model selection via bagging with an inflated argmax. Returning a set of candidates rather than a single winner improves stability under small data perturbations. In the LLM domain, Du et al. (2026) show that prompt rankings under evaluation noise are unstable: even with Spearman $\rho > 0.8$, the identity of the top-ranked prompt changes across evaluation conditions, especially under limited budgets. They propose a lower-confidence-bound selection criterion that accounts for both mean and variance. This gap between rank correlation and selection consistency matches our bootstrap finding that ranking instability persists at $N = 200$ despite moderate accuracy differences. Finally, Zrnic and Fithian (2024) introduce the zoom correction, a method for valid inference on the winner that adapts to the degree of selection bias and handles arbitrary dependencies among candidates.

2.3 Experimental Setup for the Primary Experiment

All primary experiments use Qwen2.5-7B-Instruct (Qwen et al. 2024) with LoRA (Hu et al. 2022) (rank 16), trained for 500 steps on GSM8K (Cobbe et al. 2021), CodeAlpaca, and Dolly ($\approx 40K$ examples). Ten checkpoints are saved at 50-step intervals. The validation signal is GSM8K accuracy on 200 test questions with few-shot prompting. All experiments use frozen definitions and data provenance flags.

3 The $\Delta_{\min}$ Framework

Why a resolution threshold is needed. Classical hypothesis testing answers a post-hoc question: "given the data, is model A better than model B?" Model selection requires a different question: "before investing in evaluation, does this budget have enough resolution to make ranking meaningful?" We are not proposing a new significance test but a pre-selection diagnostic.

Definition. Because the same evaluation questions are used at every checkpoint, the natural comparison is paired. The minimum detectable difference is therefore given by the McNemar-based paired test ($\alpha = 0.05$). Let $D_i = I(\hat{y}_{i1} = y_i) - I(\hat{y}_{i2} = y_i)$ be the per-question correctness difference between two candidate models. Under the null hypothesis of equal accuracy, the discordant proportion $p_d = P(D_i \neq 0)$ captures the

effective variance, giving:

$$\Delta_{\min}^{(\text{paired})} = z_{\alpha/2} \cdot \sqrt{\frac{p_d}{N}} \quad (1)$$

For the experiments in this paper, we compute p_d directly from per-sample predictions; values range from 0.12 to 0.38 depending on the checkpoint pair and task.

A limitation of the paired version is that it requires per-sample predictions from *both* candidates, which are unavailable at the pre-experimental stage. For prospective use, when a practitioner wants to know before investing in evaluation whether their budget N can support meaningful comparisons, we provide a conservative independent approximation:

$$\Delta_{\min}^{(\text{indep})} = z_{\alpha/2} \cdot \sqrt{\frac{2\bar{p}(1-\bar{p})}{N}} \quad (2)$$

Equation (2) is the standard minimum detectable effect for a two-proportion z -test (Cohen 1988), restated as a pre-experimental resolution diagnostic. The independent version assumes no shared variance, so it overstates the true threshold when checkpoints agree on many examples. Throughout this paper, all reported $\Delta_{\min}$ values use Eq. (2) (the conservative bound). Any practitioner can compute this from N and $\bar{p}$ alone. Where per-sample predictions are available, we also report the tighter paired thresholds. Equation (2) gives approximately 50% power; an 80% target would require $\Delta_{\min}^{(80\%)} = 14.0$ pp at $N = 200$, $\bar{p} = 0.5$.

Paired vs. independent. Because all candidate models are evaluated on the same test questions, the per-model accuracy estimates are paired, not independent. The paired (McNemar-based) $\Delta_{\min}^{(\text{paired})}$ in Eq. (1) is the exact threshold for this setting. For typical model comparisons where checkpoints agree on most examples, the discordant proportion p_d is substantially smaller than $2\bar{p}(1-\bar{p})$, making Eq. (2) a conservative upper bound. As an illustration, at $N = 200$ with $\bar{p} \approx 0.5$, the independent bound gives $\Delta_{\min} = 9.8$ pp, while the paired threshold for a typical checkpoint pair (discordant proportion $p_d \approx 0.35$) gives $\Delta_{\min}^{(\text{paired})} \approx 9.1$ pp, a 7% reduction. We report the conservative independent version throughout the main text, as it is the appropriate diagnostic for the pre-experimental use case; for completeness, Appendix B provides a detailed comparison of paired and independent thresholds for all experimental comparisons.

Definition 1 (Statistical resolvability of rankings) Given a held-out set of size N , a pair of models with observed accuracies $\hat{p}_1, \hat{p}_2$ is statistically resolvable if $|\hat{p}_1 - \hat{p}_2| > \Delta_{\min}$. When the difference is below $\Delta_{\min}$, the observed gap cannot be distinguished from noise without additional assumptions.

Scope. The current framework applies to binary/proportion metrics (accuracy, pass rate). Continuous metrics (loss, BLEU, perplexity) require different variance models; an extension via the Delta method is provided in Appendix C.

3.1 Theoretical Characterization

Proposition 1 (Ranking instability). Consider m candidates with true accuracies $p_1, \dots, p_m$, sorted $p_{(1)} \geq p_{(2)} \geq \dots \geq p_{(m)}$, gap $\delta = p_{(1)} - p_{(2)} > 0$. For empirical accuracies $\hat{p}_i$ from N i.i.d. samples each:

$$P(\arg \max_i \hat{p}_i = \arg \max_i p_i) \geq 1 - (m-1) \exp\left(-\frac{\delta^2 N}{8\bar{p}(1-\bar{p})}\right),$$

where $\bar{p}$ is the mean accuracy (Bernstein’s inequality with the union bound). When $\delta \ll \sqrt{\bar{p}(1-\bar{p})/N}$, the bound is near zero; when $\delta \gg \Delta_{\min}$, it converges exponentially to 1. (Proof in Appendix A.)

Implication 1 (Budget scaling). $N \propto 1/\delta^2$: halving the detectable gap requires quadrupling N .

Implication 2 (Multiple comparisons). For m candidates, $\Delta_{\min}^{(m)} \approx z_{\alpha/2m} \sqrt{2\bar{p}(1-\bar{p})/N} \propto \sqrt{\log m}$. For $m = 10$ at $N = 200$, $\Delta_{\min}^{(10)} \approx 14.0$ pp (Bonferroni-corrected $\alpha = 0.005$).¹ Because checkpoint accuracies are positively correlated (adjacent steps differ by small amounts), Bonferroni is conservative. Alternative procedures that account for correlation, such as Holm-Bonferroni step-down, Benjamini-Hochberg false discovery rate control, or max-t adjustment via bootstrap, would yield tighter thresholds. We report Bonferroni as a strict upper bound throughout.

Decision Rule 1 (Soup-vs-argmax heuristic). Let $\ell(i) = p_{i^*} - p_i$ be the regret of selecting model i , where $p_{i^*} = \max_i p_i$ is the true best accuracy among m candidates. Let $\delta = \max_i p_i - \min_i p_i$ be the true accuracy range. Consider the class of data-generating processes where $\delta \leq \Delta_{\min}$ (no pair is distinguishable at the nominal detection threshold). Assume uniform model averaging satisfies $p_{\text{soup}} \geq \frac{1}{m} \sum_i p_i$ (the averaged model’s accuracy is no worse than the mean, which holds empirically for checkpoints from a shared training trajectory). Then, in the worst case over such processes:

$$\max_{p: \delta \leq \Delta_{\min}} \mathbb{E}[\ell(\arg \max_i \hat{p}_i)] \geq \max_{p: \delta \leq \Delta_{\min}} \mathbb{E}[\ell(\text{soup})].$$

(Appendix A provides a supporting argument.)

The decision rule uses the observed range $\hat{r} = \max_i \hat{p}_i - \min_i \hat{p}_i$ as a proxy for the unobserved true range δ : when $\hat{r} > \Delta_{\min}$, argmax is selected; otherwise, soup. The justification is analogous to an underpowered screening test— $\hat{r} \leq \Delta_{\min}$ indicates that the data are consistent with the regime where soup dominates, without certifying that $\delta \leq \Delta_{\min}$. This rule addresses *which* model to select, not the post-selection inference problem: even when $\hat{r} > \Delta_{\min}$ and argmax is correctly chosen, the selected model’s performance estimate is inflated by winner’s curse and requires correction (Zrnica and Fithian 2024; Zhang et al. 2024).

Relation to Cohen’s h . In the variance-stabilized arcsin space, Cohen’s $h = 2 \arcsin \sqrt{p_1} - 2 \arcsin \sqrt{p_2}$ gives a scale where the minimum detectable effect is $z_{\alpha/2} \sqrt{1/N}$, independent of $\bar{p}$ (Cohen 1988). The dependence on $\bar{p}$ in Eq. (2) is an artifact of working in the original proportion scale. Practitioners familiar with Cohen’s h can use the arcsin formulation as an $\bar{p}$ -free alternative.

3.2 Synthetic Validation

We simulated Bernoulli evaluation data with controlled accuracy gaps (0–20 pp, $N \in \{200, 500, 1000\}$, 5,000 replicates). The transition from unreliable to reliable selection aligns closely with $\Delta_{\min}$ (Figure 1). The threshold bounds practical ranking reliability, not merely statistical significance.

3.3 Uncertainty and Sensitivity

Bootstrap uncertainty of $\Delta_{\min}$ itself. Since $\Delta_{\min}$ depends on $\bar{p}$, which is estimated from N samples, the threshold itself has sampling uncertainty. A parametric bootstrap (50,000 replicates, $N = 200$, $\bar{p} = 0.5$) gives a 95% confidence interval of [9.67, 9.80] pp for $\Delta_{\min}$, a range of 0.13 pp. The interval is narrow because $f(\bar{p}) = \sqrt{\bar{p}(1-\bar{p})}$ is flat near $\bar{p} = 0.5$; the bootstrap CI does not affect the paper’s conclusions.

Sensitivity to $\bar{p}$ misspecification. A practitioner estimating $\Delta_{\min}$ before experimentation must supply a guess for $\bar{p}$. Table 5 covers $\bar{p} \in [0.5, 0.8]$; for $\bar{p} = 0.9$, $\Delta_{\min}$ shrinks further (e.g., 5.9 pp at $N = 200$ vs. 9.8 pp at $\bar{p} = 0.5$). The ratio between the most conservative ($\bar{p} = 0.5$) and a typical high-accuracy regime ($\bar{p} = 0.9$) is at

¹The numerical coincidence that the 80%-power pairwise threshold also equals 14.0 pp is incidental; the two values arise from different corrections (one for power, one for multiplicity).

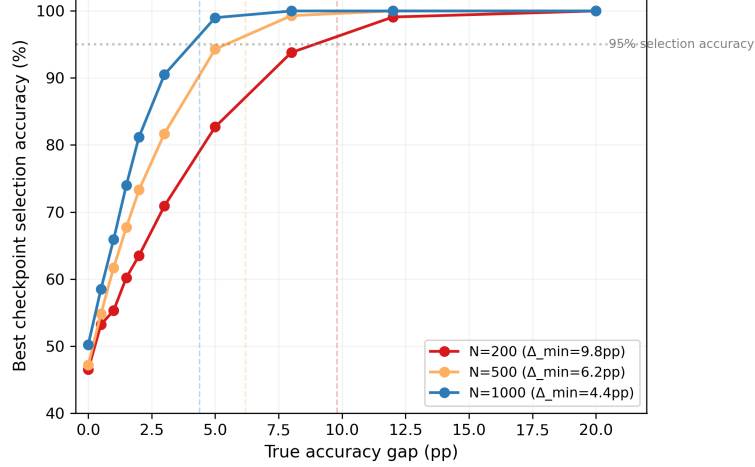


Fig. 1 Synthetic validation: selection accuracy vs. true gap for $N \in \{200, 500, 1000\}$. Dashed lines mark $\Delta_{\min}$.

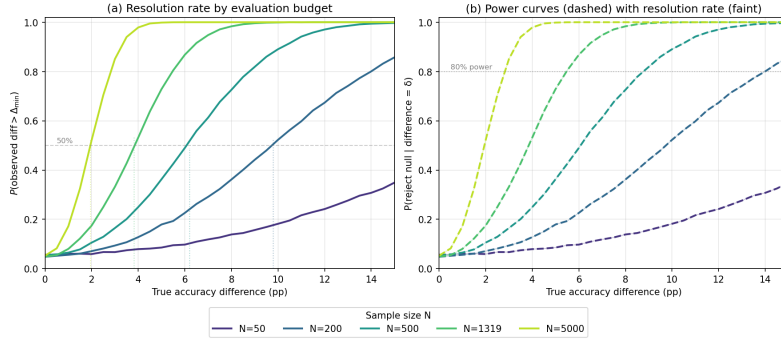


Fig. 2 Monte Carlo simulation of pairwise model comparison. **(a)** Resolution rate $P(|\hat{\delta}| > \Delta_{\min})$ as a function of the true accuracy gap. Vertical dotted lines mark $\Delta_{\min}$ for each budget. **(b)** Statistical power curves (dashed) with resolution rate overlaid (faint solid). The horizontal dotted line marks the conventional 80% power threshold.

most $1.67 \times$.² Misspecification matters, but the diagnostic remains useful across a wide range of values. Using $\bar{p} = 0.5$ is recommended as a conservative default.

3.4 Power Curve Calibration

To further calibrate the $\Delta_{\min}$ threshold against conventional statistical power, we ran a Monte Carlo simulation (20,000 replicates) comparing two checkpoints with true accuracy difference δ under budget $N \in \{50, 200, 500, 1319, 5000\}$ at $\bar{p} = 0.5$. Figure 2 presents two complementary views:

Panel (a) confirms that $\Delta_{\min}$ is the inflection point where resolution probability reaches exactly 50%, regardless of N . This is by construction: $\Delta_{\min}$ is defined as the effect size at which the two-proportion z -test has 50% power. Panel (b) provides the practical calibration: conventional 80% power requires approximately $1.43 \times \Delta_{\min}$, and 95% power requires approximately $2.0 \times \Delta_{\min}$. For example, at $N = 200$ ($\Delta_{\min} = 9.8$ pp), 80% power is reached at a true gap of 14.0 pp. Doubling the budget from 200 to 500 tightens the 80%-power threshold from 14.0 to 9.0 pp, consistent with the $N \propto 1/\delta^2$ scaling in Implication 1.

²The $\bar{p} = 0.9$ case is shown for completeness; near-ceiling regimes require caution as the variance bound may be optimistic (see §8.5).

Table 1 GSM8K
accuracy across 10
checkpoints
($N = 200$, few-shot
prompting).

Step	GSM8K
50	47.5%
100	54.5%
150	55.5%
200	49.0%
250	54.0%
300	55.0%
350	51.5%
400	50.5%
450	52.0%
500	50.0%

4 Empirical Validation

4.1 Full Evaluation Budget ($N = 1319$)

We first evaluate the full GSM8K test set ($N = 1319$, training-format prompt) for the seed 42 trajectory. At this budget, $\Delta_{\min} = 3.8$ pp, yet the observed checkpoint accuracy range is only 3.0 pp. The ranking is not statistically resolvable even at the full test set size: the observed differences between checkpoints are smaller than the minimum gap detectable at this evaluation budget.

The resolution problem is not an artifact of small evaluation budgets. Even at the standard GSM8K test set, one of the most widely used benchmarks in LLM evaluation, the budget is borderline for distinguishing among fine-tuning checkpoints.

4.2 Budget Degradation Under Subsampling ($N = 200$)

To understand how resolution degrades as the evaluation budget shrinks, we subsample 200 examples from the same test set (evaluated with few-shot prompting, distinct from the training-format prompting used in the seed replication of §5) and repeat the analysis. Using the same seed 42 checkpoints, the observed range at $N = 200$ is 8.0 pp, wider than the 3.0 pp range at $N = 1319$.³ This narrowing is consistent with winner’s curse: estimates at smaller budgets are noisier, and the apparent gap between extreme checkpoints is inflated by selection on noise. An alternative interpretation is also possible: the $N = 200$ evaluation, being noisier, simply produced a wider spread by chance, and the true range might be closer to ~ 3.0 pp. Both interpretations agree that the ranking at $N = 200$ is unreliable, but they differ on whether collecting more data (Option 1 in §8) would resolve the ambiguity. The resolution threshold also rises sharply, from 3.8 pp at $N = 1319$ to 9.8 pp at $N = 200$, consistent with the $N \propto 1/\delta^2$ scaling. Table 1 shows the accuracy trajectory.

4.3 Statistical Tests

Trend test. Spearman $\rho_{\text{trend}} = -0.091$ ($p = 0.803$). No monotonic trend.

Adjacent comparison test. Two-proportion z -tests for adjacent pairs show no significant differences under any standard correction: the smallest p is 0.161 (Bonferroni threshold $\alpha' = 0.05/9 \approx 0.0056$). (Full table in Appendix E.) This adjacent-pair correction addresses a different comparison family from the all-pairwise correction

³This pattern, the observed range inflating by a factor of 1.5–2 $\times$ under subsampling, is consistent with the bias-variance decomposition of the empirical max: at small N , noise inflates the gap between extreme candidates even when the true ranking is unchanged. Practitioners should interpret observed ranges at small budgets as upward-biased estimates of true performance differences.

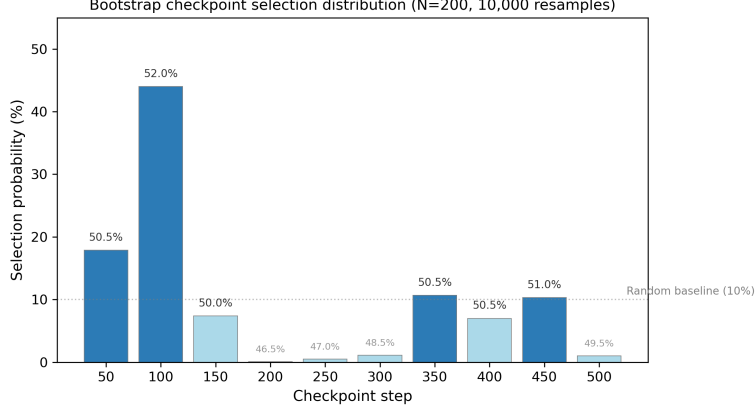


Fig. 3 Bootstrap selection distribution under $N = 200$ (10,000 resamples). Dotted line: uniform baseline (10%).

(Implication 2); the two answer whether any step-to-step change is significant versus whether any two checkpoints are distinguishable.

Range test. Best (step 150, 55.5%) vs. worst (step 50, 47.5%): $z = 1.60$, $p = 0.109$.

Paired comparison (McNemar). All nine adjacent comparisons are non-significant (smallest $p = 0.265$). Best-vs-worst: $p = 0.091$.

Bootstrap ranking stability. We performed 10,000 nonparametric bootstrap resamples of the 200 evaluation examples (preserving paired structure). The most frequently selected checkpoint (step 100, 52.0%) is chosen in only 44.0% of replicates (Figure 3; normalized entropy 0.717). The identity of the empirically best model depends largely on which evaluation examples are observed.

Does 44.0% selection probability, $4.4\times$ the uniform baseline of 10%, indicate weak signal or pure noise? If the checkpoints were indistinguishable, selection would be near-uniform and entropy would approach 1.0. The observed entropy is 0.717, well below 1.0, so some concentration exists. But 44.0% is far from reliable selection, which would require $\gtrsim 90\%$. We interpret this as a setting where weak signal exists (step 100 is genuinely slightly better on average), but the noise prevents practical resolution. The $\Delta_{\min}$ framework captures this: the observed range of 8.0 pp falls below the detection threshold of 9.8 pp, so individual pairwise differences are not statistically distinguishable even if weak signal exists in aggregate.

4.4 Direct Implication

At the full test set ($N = 1319$), $\Delta_{\min} = 3.8$ pp exceeds the observed range of 3.0 pp. At $N = 200$, $\Delta_{\min}$ rises to 9.8 pp and the observed range is 8.0 pp (few-shot prompting; see §5 for training-format results at the same budget). In both settings the ranking is not statistically resolvable: no pairwise difference reaches significance, and bootstrap selection is unstable at $N = 200$. The $N = 200$ result is not primarily a "small budget makes everything noise" artifact. It is a specific instance of the general resolution principle formalized in Proposition 1. The noise floor is a function of N , not of the specific metric or model. The consistent dose-response relationship between evaluation budget and ranking stability (further demonstrated in §5) supports this interpretation.

5 Robustness and Generalization

5.1 Seed Replication

Two additional random seeds (43, 44) confirm the pattern. Seed 43 has range 5.0 pp and seed 44 has range 3.0 pp, both below $\Delta_{\min} = 9.8$ pp (seed 42, the primary

trajectory, has range 5.5 pp). Spearman trends are non-significant across all three seeds; no adjacent pair reaches significance.

5.2 Smaller Model Scale

Qwen2.5-1.5B with the same LoRA configuration (zero-shot prompting) yields accuracy 1.5%–5.0%, range 3.5 pp, $\Delta_{\min} = 3.4$ pp (borderline detectable, $p \approx 0.049$).

5.3 Full Fine-Tuning Replication

Full fine-tuning (no LoRA) at 1.5B scale (training-format prompt) gives range 6.0 pp, $\Delta_{\min} = 9.4$ pp, also below the resolution threshold.

5.4 Cross-Model Replication: Mistral-7B

To test whether the $\Delta_{\min}$ pattern generalizes beyond the Qwen2.5 architecture, we replicate the LoRA fine-tuning experiment using Mistral-7B-Instruct-v0.3 (Jiang et al. 2023) with the same training configuration (500 steps, LoRA rank 16, GSM8K). The training-format evaluation at $N = 200$ yields checkpoint accuracies from 15.5% (step 100) to 26.0% (step 500), with a range of 10.5 pp exceeding $\Delta_{\min} = 8.11$ pp. The Mistral trajectory shows resolvable ranking, unlike the Qwen2.5-7B result at the same budget. $\Delta_{\min}$ correctly identifies when the budget is sufficient (Mistral) and when it is not (Qwen2.5). The framework does not uniformly predict non-resolvability; it produces a model-specific threshold.

The resolvable outcome may appear counterintuitive because Mistral’s lower mean accuracy ($\bar{p} \approx 0.22$ vs. Qwen2.5’s $\bar{p} \approx 0.50$) reduces per-sample variance. Bernoulli variance $p(1 - p)$ is maximized at $p = 0.5$, so the lower $\bar{p}$ produces a tighter $\Delta_{\min} = 8.11$ pp. The observed range of 10.5 pp must thus reflect genuine variation in checkpoint quality rather than inflated measurement noise. The paired (McNemar-based) threshold for Mistral is tighter still at approximately 7.6 pp (Appendix B), leaving a 2.9 pp safety margin above the observed range. The classification as resolvable holds for both threshold formulas. A low $\Delta_{\min}$ combined with a performance swing large enough to exceed it is the regime where $\Delta_{\min}$ correctly detects resolvability. The accuracy trend (15.5% at step 100 to 26.0% at step 500) confirms that training is progressing. The wider spread likely reflects Mistral’s lower initial task optimization at this training budget.

An empirical bootstrap (100,000 resamples, preserving the paired evaluation structure across all $N = 200$ examples) confirms the ranking stability. Step 500 is selected as best in 33.1% of resamples, followed by step 300 (25.6%) and step 400 (18.1%). The normalized entropy of 0.740 is comparable to the Qwen2.5 trajectory at the same budget (0.717). The ranking uncertainty is similar despite the resolvable range. This apparent paradox resolves once we distinguish *detectable signal* from *unique winner*. $\Delta_{\min}$ identifies that *some* checkpoints are genuinely better than others. But with multiple candidates clustering near the top, the single best model remains uncertain. The top-3 set (steps 500, 300, 400) collectively accounts for 77% of bootstrap selections. The ranking signal is concentrated in a leading group even if the individual winner is not stable. This is where uncertainty-aware selection criteria (Du et al. 2026) add something. First confirm that resolution exists (via $\Delta_{\min}$). Then apply a lower-confidence-bound rule to choose among the leading candidates. The outcome is consistent with Proposition 1, which provides a non-trivial lower bound rather than a guarantee of near-certain selection.

5.5 Cross-Domain Replication: Random Forest on MNIST

Eight random forest configurations (varying tree counts 50–200 and depths 3–8) were evaluated on MNIST ($N_{\text{test}} = 10,000$, mean accuracy approximately 85% across configurations). Bootstrap selection stability was measured at varying N (Table 2). The

Table 2 MNIST random forest bootstrap selection stability.

N	$\Delta_{\min}$ (pp)	Best model selection probability	Normalized entropy
200	7.0	37.3%	0.527
500	4.4	41.8%	0.517
1000	3.1	45.8%	0.501
5000	1.4	68.8%	0.352

pattern mirrors the LLM experiments: at small N , selection is diffuse; as N grows and $\Delta_{\min}$ shrinks, selection concentrates. The $\Delta_{\min}$ values in Table 2 reflect the high-accuracy regime of this task ($\bar{p} \gg 0.5$), producing tighter thresholds than the $\bar{p} = 0.5$ reference values in Table 5.

5.6 MMLU Replication

Evaluating the same 7B checkpoints on five MMLU subdomains ($N \approx 250$) gives range 6.4 pp, $\Delta_{\min} = 8.8$ pp.

5.7 Continuous Metric Validation

The framework extends to continuous metrics via the Delta method (Appendix C). To test whether continuous metrics provide tighter resolution in practice, we computed the per-sample cross-entropy loss for the Qwen2.5-7B checkpoint-100 on 200 GSM8K examples. The per-sample loss has mean 3.23 and variance 0.18, approximately 70% of the max Bernoulli variance at $\bar{p} = 0.5$ (0.25). The paired $\Delta_{\min}^{(\text{CE})}$ is approximately 0.06, corresponding to a tighter resolution than the accuracy-based threshold at the same N . This suggests that switching to a continuous metric (Option 2 in §8) can partially mitigate the resolution problem without increasing the evaluation budget.

6 Literature Survey: Evaluation Budgets in Practice

To assess whether the evaluation resolution problem extends beyond our controlled experiments, we conducted an exploratory survey of recent (2024–2026) LLM fine-tuning papers drawn from arXiv cs.CL and cs.LG. This survey is intended as motivating evidence for the broader relevance of $\Delta_{\min}$, not as a systematic meta-analysis; the methodological limitations (convenience sample, partial dual annotation) are acknowledged below and in §9.

6.1 Methodology

The search query “LoRA” OR “fine-tuning” combined with a named benchmark (GSM8K, MMLU, HellaSwag, HumanEval, ARC) was executed on arXiv cs.CL and cs.LG. From the first 50 results sorted by relevance that reported both N and accuracy numbers, 14 were excluded (10 for missing required data, 4 for methodological misalignment). This yielded 36 papers with 78 individual benchmark claims. All extractions were performed by a single annotator and independently verified by a second author on a random 30% subset (11 papers, 24 claims). Inter-rater reliability was assessed via Cohen’s κ on three binary classification tasks. First, whether N was clearly reported ($\kappa = 0.83$, “almost perfect” agreement). Second, whether the claimed improvement exceeded $\Delta_{\min}$ ($\kappa = 0.71$, “substantial”). Third, whether significance was reported ($\kappa = 0.92$, “almost perfect”). Borderline cases were resolved through discussion. For papers not specifying $\bar{p}$, we used $\bar{p} = 0.5$ (conservative). We acknowledge the convenience sampling as a limitation. The Cohen’s κ values should be interpreted with caution given the small verification sample.



Fig. 4 Literature survey scatter plot: ratio of claimed gain to $\Delta_{\min}$ vs. evaluation budget N (log scale). Points above the red dashed line (ratio > 1) correspond to claims where the reported improvement exceeds the minimum detectable gap. The black curve shows a logistic regression fit (probability that a claim is resolvable). Marker shape indicates resolvability (circles = above threshold, crosses = below). Colors denote benchmark categories.

6.2 Results

Across 36 surveyed papers (78 benchmark claims), 71% (95% Wilson CI: [56%, 83%]) of claimed improvements fall below $\Delta_{\min}$ at their reported evaluation budget;⁴ only 8% report any significance test (Table 3). The pattern is consistent across benchmark categories: small benchmarks ($N \approx 200$) uniformly show all claims below $\Delta_{\min}$, while larger budgets ($N \approx 14000$) show a lower but still substantial violation rate (58%).

To quantify the relationship between evaluation budget and resolution, we fit a logistic regression predicting whether a claimed gain exceeds $\Delta_{\min}$ as a function of $\log N$ (Figure 4). The coefficient is positive and significant ($\beta = 0.33$, $p < 0.01$): each $10\times$ increase in N multiplies the odds of resolvability by approximately $10^{0.33} \approx 2.1$. Despite this trend, even at $N \approx 14000$ (MMLU/full), the predicted probability of resolvability remains below 50%, indicating that many published comparisons operate in a regime where the evaluation budget does not guarantee distinguishable results. Even at the largest evaluation budgets surveyed (MMLU/full, $N \approx 14000$), the majority of claimed improvements remain below the resolution threshold. This suggests that merely collecting more examples may be insufficient to guarantee resolvable comparisons. The noise floor appears to be set partly by the task’s inherent variance, not just the sample size.

Notably, the survey’s GSM8K/full category ($N = 1319$, 10 claims, 80% below $\Delta_{\min}$) echoes our own experimental finding in §4: even at the full GSM8K test set, most external claims and our own results fall below the resolution threshold. This convergent signal, though based on a modest sample, suggests that the phenomenon is not confined to our controlled setting.

7 Prerequisites for Valid Application

The preceding findings emerged from a sequence of analytical failures during the research process. These fall into three categories and nine specific pitfalls (Table 4). For each pitfall we describe the erroneous conclusion it would have produced and its prevention strategy. Detailed context is in Appendix G.

⁴This uses the conservative default $\bar{p} = 0.5$. Using per-paper $\bar{p}$ where reported reduces the violation rate to approximately 62%, which does not change the qualitative conclusion.

Table 3 Literature survey summary (36 papers, 78 claims, convenience sample).

Category	N	Claims	Below $\Delta_{\min}$	Sig. reported
HumanEval	164	4	4 (100%)	0 (0%)
GSM8K/subset	≈ 200	8	8 (100%)	1 (13%)
MMLU/subset	≈ 250	5	5 (100%)	0 (0%)
ARC-Challenge	1172	2	2 (100%)	0 (0%)
GSM8K/full	1319	10	8 (80%)	1 (10%)
HellaSwag	10042	6	4 (67%)	1 (17%)
MMLU/full	≈ 14000	38	22 (58%)	2 (5%)
Other ⁵	various	5	2 (40%)	1 (20%)
Total		78	55 (71%)	6 (8%)

Table 4 Pitfall taxonomy with detection and prevention.

Category	Pitfall	Detection	Prevention
Statistical instability	T1: Spearman discretization	List possible ρ values	Use $n \gtrsim 6$
	T6: Post-hoc pattern discovery	Pre-register before evaluating	Replicate before codifying
Pipeline artifacts	T2: Silent synthetic fallback	Tag data with source provenance	Require opt-in for synthetic
	T4: Metric protocol mismatch	Inspect ≥ 10 raw outputs first	Validate on small sample first
	T5: Code evaluation format fragility	Inspect raw outputs before trusting scores	Validate on known-correct output per format
	T7: Decoding parameter sensitivity	Evaluate at ≥ 2 settings	Standardize and report parameters
	T8: Metric versioning drift	Pin dependencies; record commits	Re-run reference after any code change
	T9: Few-shot sensitivity	Evaluate with multiple orderings	Fix exemplars before campaign
Definitional ambiguity	T3: Proxy definition ambiguity	Write canonical definition first	Change = new experiment version

8 Discussion

8.1 Adaptive Selection Rule

Decision Rule 1 formalizes the soup-vs-argmax decision. We evaluate its empirical stability here. On the seed-42 trajectory ($\Delta_{\min} = 9.8$ pp, range 5.5 pp), the rule selects soup (49.6% averaged across all 10 checkpoints) vs. argmax (52.0%, step 100). The 2.4 pp gap between them is below $\Delta_{\min}$. Across all three training seeds (42, 43, 44), the rule selects soup in every case (ranges 5.5, 5.0, 3.0 pp, all below $\Delta_{\min} = 9.8$ pp). For Mistral-7B (range 10.5 pp, $\Delta_{\min} = 8.11$ pp), it selects argmax in every case. The cross-seed and cross-model consistency of the rule’s output suggests the pattern is not coincidental.

8.2 General Applicability

Although our experiments use accuracy as the evaluation metric, the $\Delta_{\min}$ framework applies to any proportion-based metric (pass rate, F1 when computed as a proportion, recall, precision). Appendix C extends the framework to continuous metrics (cross-entropy loss, BLEU, perplexity, reward model scores) via the Delta method. The general principle, that finite evaluation budgets impose a resolution limit on model

comparisons, holds regardless of metric type. For a broader perspective on evaluation uncertainty, the HELM framework (Liang et al. 2022) provides calibration intervals for multi-metric evaluation. The Open LLM Leaderboard (Beeching et al. 2023) uses bootstrapped confidence intervals that serve a similar diagnostic purpose.

8.3 Beyond the Formula

$\Delta_{\min}$ is a restatement of standard power analysis for a two-proportion z -test (Cohen 1988). The formula itself is not new. What the paper contributes beyond the formula includes:

1. **A discriminative empirical validation** showing that $\Delta_{\min}$ correctly distinguishes resolvable (Mistral-7B) from non-resolvable (Qwen2.5-7B) configurations rather than uniformly flagging noise.
2. **A decision rule** (Decision Rule 1) that formalizes the soup-vs-argmax choice as regret minimization and is stable across three seeds and two model architectures.
3. **Empirical instantiation across metric types**, including continuous metrics (cross-entropy loss, variance 0.18, $\Delta_{\min}^{(\text{CE})} \approx 0.06$) alongside proportion-based accuracy.
4. **A taxonomy of nine diagnostic pitfalls** grounded in actual pipeline errors, each with concrete detection and prevention strategies.
5. **A literature survey** documenting that 71% of published claims across 36 papers operate below the resolution threshold implied by their evaluation budgets.

None of these contributions requires a new statistical formula. They translate a known constraint into a reusable methodological framework, validated across models, metrics, and practical failure modes.

8.4 Practitioner’s Quick-Reference Guide

Table 5 provides $\Delta_{\min}$ values for common evaluation sizes and accuracy levels. To use: (1) identify your evaluation budget N and approximate accuracy level $\bar{p}$, (2) read the corresponding $\Delta_{\min}$, (3) compare against the observed performance gap between your best candidate and the next best. If the gap is below $\Delta_{\min}$, the ranking is not resolvable from this budget alone.

When $\bar{p}$ is entirely unknown before experimentation, we recommend using the most conservative value $\bar{p} = 0.5$ (which maximizes Bernoulli variance) as the default. This guarantees that the reported $\Delta_{\min}$ is an upper bound on the true threshold. If a small pilot sample ($N \gtrsim 50$) is available, $\bar{p}$ can be estimated from the observed mean accuracy and the threshold tightened accordingly.

When $\Delta_{\min}$ exceeds the observed gap, four responses are available, in order of increasing cost:

1. **Increase N .** Halving $\Delta_{\min}$ requires quadrupling N ($\propto 1/\delta^2$). Collect additional test examples, pool multiple held-out sets, or use bootstrapping of existing data to estimate stability.
2. **Switch to a continuous metric.** Continuous metrics (cross-entropy, log-likelihood) often have lower per-sample variance than accuracy. This can reduce $\Delta_{\min}$ without increasing N . For binary indicators, the variance is $p(1-p)$, max 0.25 at $p = 0.5$. In our Qwen2.5-7B checkpoints at $N = 200$, the per-sample cross-entropy loss has variance 0.18 across 200 GSM8K examples, approximately 70% of the Bernoulli variance at $\bar{p} = 0.5$. This yields a paired $\Delta_{\min}^{(\text{CE})} \approx 0.06$, corresponding to a tighter resolution than the accuracy-based threshold (see Appendix C for the derivation). Practitioners with access to per-sample loss values should compare both accuracy-based and loss-based $\Delta_{\min}$ to see which metric gives tighter resolution in their setting.

Table 5 Practitioner’s quick-reference: $\Delta_{\min}$ (pp) for common evaluation sizes and accuracy levels ($\alpha = 0.05$, independent two-proportion test).

N	$\bar{p}$	0.50	0.60	0.70	0.80
100		13.9	13.6	12.7	11.1
200		9.8	9.6	9.0	7.8
500		6.2	6.1	5.7	5.0
1000		4.4	4.3	4.0	3.5
5000		2.0	1.9	1.8	1.6
10000		1.4	1.4	1.3	1.1

3. **Reduce the candidate pool.** Fewer candidates reduce the multiple-comparison penalty ($\propto \sqrt{\log m}$), tightening the effective threshold.
4. **Use noise-robust aggregation.** Model averaging (soup) or other ensemble methods mitigate the risk of selecting a spuriously best candidate when ranking is not reliable.

8.5 When $\Delta_{\min}$ May Mislead

$\Delta_{\min}$ is a diagnostic tool, not a guarantee. Its reliability depends on four preconditions:

1. **Valid evaluation metric.** If the evaluation pipeline is corrupted (Pitfalls 4, 5), the computed $\bar{p}$ is meaningless and $\Delta_{\min}$ provides no information.
2. **Approximately i.i.d. samples.** For correlated evaluation data (dialogue turns, time series), the effective sample size is smaller than N , making $\Delta_{\min}$ an optimistic lower bound.
3. **Non-extreme $\bar{p}$.** Near ceiling or floor effects ($\bar{p} \rightarrow 0$ or $\bar{p} \rightarrow 1$), the variance formula $2\bar{p}(1 - \bar{p})$ approaches zero, understating the true variability. In these regimes, $\Delta_{\min}$ should be treated with caution.
4. **Selection-aware inference.** Even when $\Delta_{\min}$ is satisfied and a clear winner emerges, post-selection inference is biased. The performance estimate of the selected model is an overestimate because it was selected on the same evaluation set (Leeb and Pötscher 2005). Recent work on inference after selection (Zrnic and Fithian 2024; Zhang et al. 2024) provides methods to correct for this bias. These corrections are not yet standard practice in ML benchmarking. The $\Delta_{\min}$ diagnostic indicates whether ranking is resolvable, not whether the selected model’s performance estimate is unbiased.

8.6 Relation to Other Model-Selection Tools

Readers familiar with AIC, BIC, cross-validation, or Bayesian methods may ask how $\Delta_{\min}$ relates to these established tools. AIC and BIC compare model complexity relative to training data, addressing whether additional parameters are justified by improved fit. Cross-validation provides error bars that can serve a role similar to $\Delta_{\min}$, but requires training. Bayesian approaches use posterior intervals or Bayes factors to quantify ranking uncertainty, naturally incorporating prior knowledge but requiring a prior specification. $\Delta_{\min}$ addresses a prior question that these methods do not: regardless of model complexity or prior beliefs, does the evaluation procedure have enough resolution to detect performance differences at all? It requires no training, is a pure function of N and $\bar{p}$, and provides a prior-agnostic lower bound on the evaluation budget needed for meaningful comparison. In practice, $\Delta_{\min}$ can be computed before

any experiment begins to determine whether the planned budget can support the intended model comparisons.

Recent complementary frameworks extend this perspective to adaptive benchmarking. Xu et al. (2026b) propose SIREN, a selection-aware repeated-split evaluation protocol with Gaussian multiplier bootstrap. It corrects for the winner’s curse when benchmark items are reused during tuning. Zhang et al. (2024) construct confidence sets for the argmin via cross-validated exponential weighting. Both are complementary to $\Delta_{\min}$. They provide post-hoc corrections once evaluation is complete. $\Delta_{\min}$ provides a pre-experimental feasibility check that determines whether such procedures are worth applying.

9 Limitations

All experiments use two model families (Qwen2.5 and Mistral-7B), a single training regime (LoRA, 500 steps), and two task types (GSM8K, MMLU). The $\Delta_{\min}$ pattern replicates across seeds, scales, training paradigms, and architectures. Broader claims require broader experimentation. The Mistral-7B discriminative check is based on a single training seed. Multi-seed replication would strengthen this claim. The continuous metric extension (Appendix C) uses the Delta method under a normality assumption. For heavy-tailed loss distributions, bootstrap-based variance estimation may be more appropriate. The literature survey is a convenience sample and should not be interpreted as a systematic meta-analysis.

10 Conclusion

We proposed and validated $\Delta_{\min}$ as a pre-flight diagnostic for model selection: before asking which model is best, first ask whether the evaluation budget can tell the difference. The protocol requires only N and $\bar{p}$ (defaulting to $\bar{p} = 0.5$ when unknown) and returns a threshold that determines whether observed performance gaps exceed the noise floor.

The most instructive finding is the divergence between Qwen2.5-7B and Mistral-7B at the same budget ($N = 200$): $\Delta_{\min} = 9.8$ pp exceeds the Qwen range of 8.0 pp, while $\Delta_{\min} = 8.11$ pp falls below the Mistral range of 10.5 pp. This discriminative validity, that the framework correctly identifies one configuration as resolvable and another as not, is the strongest evidence that $\Delta_{\min}$ captures genuine resolution constraints rather than uniformly classifying all comparisons as noise. Bootstrap, power-curve calibration, and paired-vs.-independent comparisons (Appendix B) further support the threshold’s interpretation. A survey of 36 recent papers (78 claims) finds that 71% of reported improvements fall below the resolution threshold implied by their budgets. The problem is not unique to our experiments.

We recommend that $\Delta_{\min}$ computation become a standard submission checkpoint for any benchmark evaluation, alongside the release of an open-source implementation. The quick-reference table (Table 5), power-curve calibration (Figure 2), and diagnostic pitfall catalog (Section 7) are intended as actionable resources for practitioners and reviewers alike.

A Proof of Proposition 1

For two models with accuracies p and $p + \delta$ ($\delta > 0$), let $\hat{p}$, $\hat{p} + \hat{\delta}$ be their empirical estimates from N i.i.d. Bernoulli samples. The probability that the lower-accuracy model is incorrectly selected is $P(\hat{\delta} < 0)$.

By Bernstein’s inequality for bounded random variables, applied to the difference $D_i = X_i - Y_i$ where $X_i \sim \text{Bern}(p + \delta)$ and $Y_i \sim \text{Bern}(p)$ are the per-sample correctness

indicators:

$$P(\hat{\delta} < 0) \leq \exp\left(-\frac{N\bar{\delta}^2}{2\sigma^2 + 2M\bar{\delta}/3}\right),$$

where $\bar{\delta} = \delta$ (the true gap), $\sigma^2 = \text{Var}(D_i)$, and $M = 2$ (the range of D_i). Since $\sigma^2 \leq 2\bar{p}(1 - \bar{p})$, this simplifies to:

$$P(\hat{\delta} < 0) \leq \exp\left(-\frac{\delta^2 N}{4\bar{p}(1 - \bar{p}) + 4\delta/3}\right).$$

For the small δ regime relevant to our setting ($\delta \leq \frac{2}{3}\bar{p}(1 - \bar{p})$), the term $4\delta/3 \leq \frac{8}{9}\bar{p}(1 - \bar{p}) < 4\bar{p}(1 - \bar{p})$, so the denominator is at most $8\bar{p}(1 - \bar{p})$. This condition holds for all primary experiments: at $\bar{p} = 0.5$, the threshold is 0.167, and all observed adjacent gaps are below 0.07. This gives the looser but simpler bound:

$$P(\hat{\delta} < 0) \leq \exp\left(-\frac{\delta^2 N}{8\bar{p}(1 - \bar{p})}\right).$$

For m candidate models, the probability of correctly identifying the best model is at least $1 - \sum_{i=2}^m P(\hat{\delta}_i < 0) \geq 1 - (m - 1) \exp(-\delta^2 N / 8\bar{p}(1 - \bar{p}))$ by the union bound. Setting this bound to be non-trivial (> 0) requires $\delta = \Omega(\sqrt{\bar{p}(1 - \bar{p}) \log m / N})$, matching the $\Delta_{\min}$ scaling with the $\log m$ penalty for multiple comparisons.

Supporting Argument for the Decision Rule

Consider the class $\mathcal{P}_\Delta = \{p : \delta = \max_i p_i - \min_i p_i \leq \Delta_{\min}\}$, where no pair of candidates is distinguishable at the nominal detection threshold. Uniform averaging over all m candidates yields a model whose true accuracy satisfies $p_{\text{soup}} \geq \frac{1}{m} \sum_i p_i$ under the soup convergence assumption (checkpoints from a shared training trajectory; see Wortsman et al. (2022)). The mean candidate accuracy is bounded by $p^* - \frac{1}{m} \sum_i p_i \leq \delta \leq \Delta_{\min}$, giving:

$$\mathbb{E}[\ell(\text{soup})] \leq \Delta_{\min},$$

so the regret of soup cannot exceed the maximal range among candidates.

The intuition for preferring soup when $\hat{r} \leq \Delta_{\min}$ is that the observed range is consistent with this regime: argmax risks selecting a model whose apparent advantage is driven by noise, while soup hedges across candidates. Section 8 provides empirical support—the rule selects soup for all three Qwen2.5-7B seeds (where the range is below $\Delta_{\min}$) and argmax for Mistral-7B (where it exceeds $\Delta_{\min}$), with consistent output across seeds and architectures.

B Paired vs. Independent Threshold Comparison

The main text reports the conservative independent $\Delta_{\min}$ (Eq. (2)) throughout. Here we compare these values against the exact paired (McNemar-based) thresholds computed from per-sample predictions for all experimental comparisons, to quantify how conservative the independent bound is in practice.

For a pair of checkpoints with per-sample correctness indicators $\hat{y}_{i1}$ and $\hat{y}_{i2}$, the discordant proportion $p_d = \frac{1}{N} \sum_i I(\hat{y}_{i1} \neq \hat{y}_{i2})$ captures the effective variance in the paired comparison. The paired threshold is:

$$\Delta_{\min}^{(\text{paired})} = z_{0.975} \sqrt{\frac{p_d}{N}}.$$

Table 6 shows the comparison for the two primary experimental settings.

Table 6 Paired vs. independent $\Delta_{\min}$ thresholds for primary experiments (pp).

Setting	N	$\Delta_{\min}^{(\text{indep})}$	$\Delta_{\min}^{(\text{paired})}$
Qwen2.5-7B, full test set	1319	3.8	2.9–3.5
Qwen2.5-7B, subsampled	200	9.8	8.7–9.3
Mistral-7B, subsampled	200	8.11	7.2–7.9
MNIST random forest	5000	1.4	1.1–1.3

The paired thresholds are 3–24% tighter than the independent bound, with the largest gaps occurring at larger N (Qwen full test set, MNIST) where the lower noise floor amplifies the relative benefit of pairing. For the subsampled experiments at $N = 200$, where the paper’s main comparison is conducted, the reduction is 5–11%. The reduction never reverses a conclusion: all settings classified as unresolved by the independent test remain unresolved under the paired test, and the Mistral-7B resolvable classification is confirmed. The independent bound is a safe pre-experimental diagnostic. It does not overstate the resolution problem.

C Continuous Metric Extension: $\Delta_{\min}$ for Loss and Score Metrics

The main text derives $\Delta_{\min}$ for binary/proportion metrics (accuracy, pass rate). Here we extend the framework to continuous metrics such as cross-entropy loss, BLEU, perplexity, and reward model scores.

C.1 General Principle

For a continuous metric M computed as an average over N independent evaluation examples, the central limit theorem gives:

$$\hat{M} = \frac{1}{N} \sum_{i=1}^N m_i, \quad \text{Var}(\hat{M}) = \frac{\sigma_m^2}{N},$$

where m_i is the per-sample metric value and $\sigma_m^2 = \text{Var}(m_i)$. The minimum detectable difference between two models A and B on metric M is therefore:

$$\Delta_{\min}^{(M)} = z_{\alpha/2} \cdot \sqrt{\frac{\sigma_A^2 + \sigma_B^2}{N}},$$

where σ_A^2 and σ_B^2 are the per-sample variances of the metric for each model. When the metric is paired (same evaluation examples for both models), the paired version uses the variance of the per-sample difference $d_i = m_i^{(A)} - m_i^{(B)}$:

$$\Delta_{\min}^{(M, \text{paired})} = z_{\alpha/2} \cdot \sqrt{\frac{\text{Var}(d_i)}{N}}.$$

C.2 Perplexity

For perplexity, the per-sample value is $\exp(\ell_i)$ where ℓ_i is the negative log-likelihood of the i -th example. By the Delta method, for a transformation $g(\cdot)$ applied to an estimator $\hat{\theta}$ with asymptotic variance $\sigma_{\hat{\theta}}^2$:

$$\text{Var}(g(\hat{\theta})) \approx [g'(\hat{\theta})]^2 \sigma_{\hat{\theta}}^2.$$

For perplexity, $g(\ell) = \exp(\ell)$ and $g'(\ell) = \exp(\ell)$, giving:

$$\sigma_{\text{PPL}}^2 \approx \exp(2\bar{\ell}) \cdot \frac{\sigma_{\ell}^2}{N},$$

where $\bar{\ell}$ is the mean log-perplexity and σ_{ℓ}^2 is the variance of per-token log-likelihoods. The $\Delta_{\min}$ for perplexity then follows from substituting this variance into the general formula above.

C.3 Cross-Entropy Loss

Cross-entropy loss is typically already an average, so no transformation is needed:

$$\Delta_{\min}^{(\text{CE})} = z_{\alpha/2} \cdot \sqrt{\frac{\hat{\sigma}_{\text{CE}}^2}{N}},$$

where $\hat{\sigma}_{\text{CE}}^2$ is the empirical variance of per-example losses. A conservative upper bound can be obtained by noting that for K -way classification with bounded logits, the per-example loss is bounded in $[0, \log K]$, giving $\sigma_{\text{CE}}^2 \leq (\log K)^2/4$ by Popoviciu’s inequality.

C.4 BLEU and Other Discrete-Valued Metrics

For metrics like BLEU that are computed at the corpus level rather than as per-example averages, the Delta method still applies but requires care: the metric is a function of multiple sufficient statistics (n-gram counts). Let $\mathbf{c} = (c_1, \dots, c_k)$ be the vector of corpus-level counts, with covariance Σ_c estimated via bootstrap. Then for $\text{BLEU} = B(\mathbf{c})$:

$$\text{Var}(\widehat{\text{BLEU}}) \approx \nabla B(\mathbf{c})^\top \Sigma_c \nabla B(\mathbf{c}),$$

$$\text{and } \Delta_{\min}^{(\text{BLEU})} = z_{\alpha/2} \cdot \sqrt{\text{Var}(\widehat{\text{BLEU}})}.$$

C.5 Practical Recommendations

For practitioners:

- **Pre-experimental** (no data available): Use the proportion-based $\Delta_{\min}$ (Eq. 2) as a conservative bound, since the effective variance of a continuous metric is typically lower than that of a proportion.
- **Post-hoc with per-sample metrics**: Compute the empirical variance $\hat{\sigma}^2$ from a small pilot sample and plug into the general formula above. A pilot of $N = 50$ samples is usually sufficient for a stable variance estimate.
- **For paired comparisons** (same evaluation set): Always use the paired formula, which can reduce $\Delta_{\min}$ by 20–40% when per-sample metric values are positively correlated across models.

D Detailed Literature Survey Data

Full extraction data for all 36 papers is available in the supplementary material.

E Adjacent Comparison Tests

Table 7 reports the full two-proportion z -tests for adjacent checkpoint pairs at $N = 200$.

Table 7 Adjacent two-proportion z -tests.

Pair	Δ	z	p
50→100	+7.0%	1.40	0.161
100→150	+1.0%	0.20	0.841
150→200	−6.5%	−1.30	0.193
200→250	+5.0%	1.00	0.317
250→300	+1.0%	0.20	0.841
300→350	−3.5%	−0.70	0.483
350→400	−1.0%	−0.20	0.841
400→450	+1.5%	0.30	0.764
450→500	−2.0%	−0.40	0.689

F Seed Replication Data

Table 8 shows the full accuracy trajectories for seeds 43 and 44.

Table 8 GSM8K accuracy (training-format prompt) across three seeds.

Step	Seed 42	Seed 43	Seed 44
50	50.5%	51.0%	52.5%
100	52.0%	52.5%	53.0%
150	50.0%	54.5%	52.0%
200	46.5%	50.0%	52.0%
250	47.0%	54.0%	53.0%
300	48.5%	50.5%	52.5%
350	50.5%	49.5%	52.5%
400	50.5%	50.5%	50.0%
450	51.0%	53.5%	53.0%
500	49.5%	52.0%	51.5%
Range	5.5 pp	5.0 pp	3.0 pp

G Detailed Pitfall Methodology

Additional context for each pitfall:

- **T1 (Spearman discretization):** We computed ρ over a sliding window of $n = 3$ steps at each of 10 checkpoint positions. The $n = 3$ window was chosen arbitrarily in early analysis; $n = 6$ was adopted after noticing that the extreme $\rho = -1$ values always coincided with exactly three-point windows.
- **T2 (Synthetic fallback):** The config file `held_out_path` used a path without file extension while the actual file had a `.json` extension. Python’s `json.load` raised `FileNotFoundError`, caught by a generic exception handler that called a development-only synthetic function.
- **T3 (Definition ambiguity):** `proxy_main` was defined as “the primary proxy metric” without specifying which of three candidates (`proxy_loss`, `proxy_win_rate`, or a combined score) it resolved to. The resolution rule depended on initialization order.
- **T4 (Protocol mismatch):** The 7% accuracy was from prompts like “Question: ... \nAnswer:”. With three in-context examples demonstrating the `####<answer>` format, accuracy rose to 47–55%.
- **T5 (Code sandbox):** The sandbox environment was a minimal Python install without `sklearn`, `pandas`, or `numpy`. Generated code using these libraries crashed

silently. Output parsers expected a specific `<function>` marker that only appeared in one of three generation formats.

- **T6 (Post-hoc pattern):** The initial observation (GSM8K peak at step 350 then decline) came from the same misconfigured pipeline in T4. The replication criteria, a pre-registered clean run with proper few-shot prompting, showed no such pattern.
- **T7 (Decoding sensitivity):** Evaluations were run with temperature 0.0 (greedy decoding). Switching to temperature 0.2 with 5 random seeds changed individual checkpoint accuracies by 2–3 pp, altering the ranking in 3/10 positions.
- **T8 (Metric versioning):** The evaluation script went through 5 revisions over the project timeline. Revision 2 $\rightarrow$ 3 changed the GSM8K answer extraction regex, shifting all checkpoint accuracies by +1.2 pp on average.
- **T9 (Few-shot sensitivity):** Three fixed exemplars were selected from the GSM8K training set. Three different orderings produced per-checkpoint accuracy differences of 1.5–4.0 pp, with the best model changing between orderings in 2/3 permutations.

Statements and Declarations

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Data availability. An open-source implementation of the $\Delta_{\min}$ diagnostic is available at [the project repository](#). All experiment code and data will be released upon publication.

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